

Evaluation Metrics Reference Guide

MAE

↓ Lower is Better

Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Range: [0, +∞)

Average absolute difference between predicted and true scores.

Note: Penalizes all errors equally.

r

↑ Higher is Better

Pearson Correlation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}}$$

Range: [-1, +1]

Linear correlation between predicted and true scores.

Note: 1 = perfect, 0 = no linear relation.

K

↑ Higher is Better

Cohen's Weighted Kappa

$$K = \frac{P_o - P_e}{1 - P_e}$$

Range: (-∞, +1]

Agreement beyond chance; quadratic weights penalize larger disagreements. P_o =weighted observed, P_e =weighted expected.

Note: 1 = perfect, 0 = chance level.

AAUC

↑ Higher is Better

Average AUC (One-vs-Rest)

$$AAUC = \frac{1}{|T|} \sum_{t \in T} AUC(y \geq t, \hat{y} \geq t)$$

Range: [0, 1]

Mean AUC across all ordinal thresholds (one-vs-rest binarization per threshold t).

Note: 1 = perfect, 0.5 = random classifier.

WD

↓ Lower is Better

Wasserstein Distance

$$WD(P, Q) = \inf_{\gamma \in \Pi(P, Q)} E_{(x, y) \sim \gamma} [|x - y|]$$

Range: [0, +∞)

Earth Mover's Distance: minimum 'cost' to transform predicted into true distribution.

Note: 0 = identical distributions.

Direction Summary

↑ Higher is Better

r k AAUC

↓ Lower is Better

MAE WD

Scoring scale: 0 – 4 (ordinal)